

Section 1.7 - Question 19

a Proof by Contradiction:

- Assume the the negation of the conclusion is false and the hypothesis is true.
 - That is, assume $n^3 + 5$ is odd and n is odd.
- if n is odd, then it can be written as $2k + 1$ for some integer k .
- Then $n^3 + 5 = (2k + 1)^3 + 5 = 8k^3 + 12k^2 + 6k + 6 = 2(4k^3 + 6k^2 + 3k + 3)$.
- However, this is a contradiction because $n^3 + 5$ is odd and $2(4k^3 + 6k^2 + 3k + 3)$ is even.
- Therefore, the negation of the conclusion is false and the original statement is true.
- Therefore, the original statement is true.

b Proof by Contraposition:

- Assume the contrapositive: if n is odd, then n is not an integer or $n^3 + 5$ is even.
- If n is odd, then it can be written as $2k + 1$ for some integer k .
- Then $n^3 + 5 = (2k + 1)^3 + 5 = 8k^3 + 12k^2 + 6k + 6 = 2(4k^3 + 6k^2 + 3k + 3)$.
- Since $4k^3 + 6k^2 + 3k + 3$ is an integer, $n^3 + 5$ is even.
- Therefore, the contrapositive is true and the original statement is true.

Section 1.7 - Question 23

$P(n)$: "If a and b are positive real numbers, then $(a + b)^n \geq a^n + b^n$."
Prove that $P(1)$ is true. What kind of proof did you use?

$P(1)$: "If a and b are positive real numbers, then $(a + b)^1 \geq a^1 + b^1$."

Direct Proof:

- Assume a and b are positive real numbers.
- Then $(a + b)^1 = a + b$ and $a^1 + b^1 = a + b$ by way of the identity property of multiplication.
- Since $a + b = a + b$, the original statement is true.

Section 1.7 - Question 29

Prove that if n is a positive integer, then n is odd if and only if $5n + 6$ is odd.

- Part 1: Prove that if n is odd, then $5n + 6$ is odd.
 - Assume n is odd.
 - Then n can be written as $2k + 1$ for some integer k .
 - Then $5n + 6 = 5(2k + 1) + 6 = 10k + 5 + 6 = 10k + 11 = 2(5k + 5) + 1$.
 - Since $5k + 5$ is an integer, $5n + 6$ is odd.
 - Therefore, if n is odd, then $5n + 6$ is odd.
- Part 2: Prove that if $5n + 6$ is odd, then n is odd.
 - Assume n is odd and $5n + 6$ is even.
 - Then n can be written as $2k + 1$ for some integer k .
 - Then $5n + 6 = 5(2k + 1) + 6 = 10k + 5 + 6 = 10k + 11 = 2(5k + 5) + 1$.
 - However, this is a contradiction because $5n + 6$ is odd and $2(5k + 5) + 1$ is even.

Section 1.7 - Question 33

Show that these statements about the integer x are equivalent:

- (i) $x + 2$ is even
 - if $x + 2$ is even, then $x + 2 = 2k$ for some integer k .
 - Then $x = 2k - 2 = 2(k - 1)$
 - Since $k - 1$ is an integer, $2(k - 1)$ is even.
 - Therefore, if $x + 2$ is even, then x is even.
 - Since $x + 2$ is even, then x is even.
- (ii) $x + 5$ is odd
 - if $x + 5$ is odd, then $x + 5 = 2k + 1$ for some integer k .
 - Then $x = 2k + 1 - 5 = 2k - 4 = 2(k - 2)$
 - Since $k - 2$ is an integer, $2(k - 2)$ is even.
 - Therefore, if $x + 5$ is odd, then x is even.
 - Since $x + 5$ is odd, then x is even.
- (iii) x^2 is even.
 - Assume the contrapositive: if x is odd, then x^2 is odd.
 - If x is odd, then $x = 2k + 1$ for some integer k .
 - Then $x^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$.
 - Since $2k^2 + 2k$ is an integer, x^2 is odd.
 - Therefore, if x is odd, then x^2 is odd.
 - Since x^2 is even, then x is even.

Therefore, all three statements are stating that x is even.

Section 1.7 - Question 43

If these four statements are equivalent, then " n is even" $\leftrightarrow$ " $n + 1$ is odd" $\leftrightarrow$ " $3n + 1$ is odd" $\leftrightarrow$ " $3n$ is even"

- " $n + 1$ is odd" $\rightarrow$ " n is even"
 - Assume the contrapositive: if n is odd, then $n + 1$ is even.
 - If n is odd, then $n = 2k + 1$ for some integer k .
 - Then $n + 1 = 2k + 1 + 1 = 2k + 2 = 2(k + 1)$.
 - Since $k + 1$ is an integer, $2(k + 1)$ is even.
 - Therefore, if n is odd, then $n + 1$ is even.
 - Therefore, if $n + 1$ is odd, then n is even.
- " n is even" $\rightarrow$ " $n + 1$ is odd"
 - if n is even, then $n = 2k$ for some integer k .
 - Then $n + 1 = 2k + 1$.
 - Since k is an integer, $2k + 1$ is odd.
 - Therefore, if n is even, then $n + 1$ is odd.
- " $3n + 1$ is odd" $\rightarrow$ " n is even".
 - Assume the contrapositive: if n is odd, then $3n + 1$ is even.
 - If n is odd, then $n = 2k + 1$ for some integer k .
 - Then $3n + 1 = 3(2k + 1) + 1 = 6k + 3 + 1 = 6k + 4 = 2(3k + 2)$.
 - $2(3k + 2)$ is even since $3k + 2$ is an integer.
 - Therefore, if n is odd, then $3n + 1$ is even.
 - Therefore, if $3n + 1$ is odd, then n is even.
- " n is even" $\rightarrow$ " $3n + 1$ is odd".
 - if n is even, then $n = 2k$ for some integer k .
 - Then $3n + 1 = 3(2k) + 1 = 6k + 1 = 2(3k) + 1$.
 - Since $3k$ is an integer, $2(3k)$ is even.
 - Therefore, if n is even, then $3n + 1$ is odd.
- " $3n$ is even" $\rightarrow$ " n is even".
 - Assume the contrapositive: if n is odd, then $3n$ is odd.
 - If n is odd, then $n = 2k + 1$ for some integer k .
 - Then $3n = 3(2k + 1) = 6k + 3 = 2(3k + 1) + 1$.
 - Since $3k + 1$ is an integer, $2(3k + 1)$ is even. and $2(3k + 1) + 1$ is odd.
- " n is even" $\rightarrow$ " $3n$ is even".
 - if n is even, then $n = 2k$ for some integer k .
 - Then $3n = 3(2k) = 6k = 2(3k)$.
 - Since $3k$ is an integer, $2(3k)$ is even.
 - Therefore, if n is even, then $3n$ is even.

Conclusion: Since all four statements are biconditional, they are logically equivalent.

Section 1.8 - Question 5

Prove that if x and y are real numbers, then $\max(x, y) + \min(x, y) = x + y$. [Hint: Use a proof by cases, with the two cases corresponding to $x \geq y$ and $x < y$, respectively.]

- Case 1: $x \geq y$
 - Then $\max(x, y) = x$ and $\min(x, y) = y$.
 - Then $\max(x, y) + \min(x, y) = x + y$.
 - Therefore, if $x \geq y$, then $\max(x, y) + \min(x, y) = x + y$.
- Case 2: $x < y$
 - Then $\max(x, y) = y$ and $\min(x, y) = x$.
 - Then $\max(x, y) + \min(x, y) = y + x$.
 - Therefore, if $x < y$, then $\max(x, y) + \min(x, y) = x + y$.

Section 1.8 - Question 9

Prove the triangle inequality, which states that if x and y are real numbers, then $|x| + |y| \geq |x + y|$ (where $|x|$ represents the absolute value of x , which equals x if $x \geq 0$ and equals $-x$ if $x < 0$).

- Case 1: $x \geq 0$ and $y \geq 0$
 - Then $|x| = x$ and $|y| = y$.
 - Then $|x| + |y| = x + y$.
 - Then $|x + y| = x + y$.
 - Therefore, if $x \geq 0$ and $y \geq 0$, then $|x| + |y| = |x + y|$.
- Case 2: $x < 0$ and $y < 0$
 - Then $|x| = -x$ and $|y| = -y$.
 - Then $|x| + |y| = -x + (-y) = -(x + y)$.
 - Then $|x + y| = -(x + y)$.
 - Therefore, if $x < 0$ and $y < 0$, then $|x| + |y| = |x + y|$.
- Case 3: $x \geq 0 \vee y < 0$
 - Absolute value of a positive number plus a negative number is always less than or equal to the positive number.
 - On the other hand, the sum of two numbers that must be positive is always greater than or equal to either of the numbers.

Section 1.8 - Question 19

Suppose that a and b are odd integers with $a \neq b$. Show there is a unique integer c such that $|a - c| = |b - c|$.

- Possibility 1: $c \geq 0$
 - Then $|a - c| = a - c$ or $c - a$ and $|b - c| = b - c$ or $c - b$.
 - If $a - c = b - c$, then $a = b$, which is a contradiction.
 - If $a - c = c - b$, then $a + b = 2c$, so $c = (a + b)/2$.
 - Therefore, if $c \geq 0$, then there is a unique integer c such that $|a - c| = |b - c|$.

Section 1.8 - Question 21

Show that if n is an odd integer, then there is a unique integer k such that n is the sum of $k - 2$ and $k + 3$.

- If n is an odd integer, then $n = 2x + 1$ for some integer x .
- $k - 2 + k + 3 = 2k + 1$.
- Therefore, if n is an odd integer, then there is a unique integer k such that n is the sum of $k - 2$ and $k + 3$.

Section 1.8 - Question 31

Prove that there is no positive integer n such that $n^2 + n^3 = 100$.

- $n^2 + n^3 = n^2(n + 1)$.
- if $n^2(n + 1) = 100$, then $n^2 = 100/(n + 1) = \sqrt{100/(n + 1)}$
 $= 10/\sqrt{n + 1}$.
- Since n is a positive integer, $n + 1$ is a positive integer.
- Therefore, $10/\sqrt{n + 1}$ is a positive integer.
- However, $10/\sqrt{n + 1}$ because the only numbers that divide 10 are 1, 2, 5, and 10 and the only way to get 1, 2, 5, 10 from $\sqrt{n + 1}$ is if $n = 0, 24, 99$.
- n cannot be 0 because n is a positive integer.
- If $n = 24$, then $n^2 + n^3 = 24^2 + 24^3 = 576 + 13824 = 14300$.
- If $n = 99$, then $n^2 + n^3 = 99^2 + 99^3 = 9801 + 970299 = 980100$.

Section 1.8 - Question 47

Use a proof by exhaustion to show that a tiling using dominoes of a 4×4 checkerboard with opposite corners removed does not exist. [Hint: First show that you can assume that the squares in the upper left and lower right corners are removed. Number the squares of the original checkerboard from 1 to 16, starting in the first row, moving right in this row, then starting in the leftmost square in the second row and moving right, and so on. Remove squares 1 and 16. To begin the proof, note that square 2 is covered either by a domino laid horizontally, which covers squares 2 and 3, or vertically, which covers squares 2 and 6. Consider each of these cases separately, and work through all the subcases that arise.]

To prove that a tiling using dominoes of a 4×4 checkerboard with opposite corners removed does not exist:

- Assume squares 1 and 16 are removed.
- Start with square 2:
 - If covered horizontally: covers squares 2 and 3. Impossible to cover remaining squares.
 - If covered vertically: covers squares 2 and 6. Impossible to cover remaining squares.